TARUN JOSEPH

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SUMMARY

MEng candidate in Industrial Engineering (Data Analytics & Machine Learning) at the University of Toronto, graduating December 2026. At SafeAlign AI, through the Vector Institute, I build runtime monitoring for multi-agent AI: fine-tuning small language models, auditing evaluation data and mapping threats to governance controls. Before graduate school I built data, ML and generative AI systems at Tata Consultancy Services, and I now teach programming and machine learning as a graduate teaching assistant.

FOCUS AREAS

Runtime security of multi-agent AI systems · Calibrated detection and escalation · Containment of tool-using agents · Evaluation methodology for agentic AI · Governance evidence and traceability

EDUCATION

Master of Engineering, Industrial Engineering Sep 2025 – Dec 2026 (expected)
University of Toronto, Canada

- Emphasis in Data Analytics & Machine Learning; Entrepreneurship and Management (ELITE). **GPA 3.8/4.0.**
- Courses: MIE1626 Data Science Methods and Statistical Learning (**A+**), MIE1517 Deep Learning (**A**), APS1070 Foundations of Data Analytics and ML (**A**), APS1080 Reinforcement Learning, APS1052 AI in Finance.
- In progress: ECE1786 Applications of Natural Language Processing, MIE1513 Decision Support Systems.

Bachelor of Technology, Industrial Engineering Aug 2017 – Jun 2021
College of Engineering Trivandrum (APJ Abdul Kalam Technological University), India

- Relevant courses: Operations Research; Data Analysis using R and Python.

AI AND RESEARCH EXPERIENCE

Machine Learning Associate, Project OSPREY Aug 2026 – Present
Vector Institute FastLane × SafeAlign AI, Toronto

- Built a **runtime monitoring pipeline** for multi-agent AI: SLMs screen user messages and classify tool outputs, OPA rules check each tool call, and every step gets a PASS, ESCALATE or QUARANTINE verdict.
- Fine-tuned a **small language model** with QLoRA on an NVIDIA DGX Spark to name the kind of attack in a flagged tool output, tested on attacker goals unseen in training against a simple baseline.
- Fine-tuned a second small model as a **user-prompt screen** for injection attempts, small enough to run on-premises.
- Found **label leakage** in the original telemetry data and rebuilt it as twin-paired sessions evaluated with grouped cross-validation.
- Built training sets with an **LLM labeller**, with no attacker goal or customer shared between training and test.
- Built the **threat taxonomy and crosswalk**: merged four dataset vocabularies into one taxonomy and linked each threat to controls, regulations, OWASP and MITRE ATLAS.
- Benchmarked **BM25 against embedding retrieval** for linking each verdict to a client's own policy documents.

MineMentor: When Should an AI Tutor Help? Oct 2026 – Present
Course research project (ECE1786), University of Toronto

- Multi-agent Minecraft tutor that decides when to hint, demonstrate or let the player struggle; a planned pilot compares it with an always-helpful companion.
- I own **calibration, data labelling, evaluation and testing** in a team of two.

- Designed the calibration protocol: play is cut into 30-second windows labelled blind from screen recordings, with **Cohen's kappa** between labellers.
- Detector thresholds are tuned by grid search on some sessions and scored once on **held-out sessions** against an LLM labeller and an idle-time baseline.

AI Program Design and Delivery Intern May 2026 – Aug 2026

Vector Institute, Toronto

- Helped lead delivery of the **Agentic AI Forecasting & Evaluation bootcamp** (9.0/10 rating) for industry teams from CIBC, PwC and Roche.
- Delivered 3 bootcamps, an applied AI cohort, an ICML 2026 speaker series and an AI ROI roundtable.
- Automated the bootcamp lifecycle with the Slack API and Apps Script, saving 5+ hours per bootcamp.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Jan 2023 – May 2025

Tata Consultancy Services (TCS), India

- Deployed GPT-3.5 and Llama 3 **RAG chatbots and AI agents** on a learning platform used by 600,000 employees.
- Took a **support-ticket agent from proof of concept to MVP**, with tool calling and vector-store memory, saving each developer 5+ hours a week.
- Built a web-scraping and OCR tool for data extraction from documents and web pages.
- Designed AI/ML coding assessments taken by 10,000+ employees each quarter, used for promotion decisions.
- Organized weekly and monthly AI/ML hackathons and coding competitions that raised program registrations.

Data Engineer May 2022 – Dec 2022

Tata Consultancy Services (TCS), India

- Owned data reporting for the talent-development senior engineer program, from database to front-end portal.
- Built PL/SQL tables and pipelines, Informatica ETL and pgAdmin monitoring; delivered 30+ user stories.
- Gathered requirements with business analysts; earned a customer satisfaction rating of 8 (FY23 Q3).

Data Analyst Jan 2022 – Apr 2022

Tata Consultancy Services (TCS), India

- Built a SQL database, Tableau dashboard and Python mailers, cutting manual effort by about 50%.
- Analysis of enrollment and completion data informed program changes that lifted learning hours by 11% (FY22).
- Received the TCS On the Spot Award for insights presented to senior management.

TEACHING EXPERIENCE

Teaching Assistant Sep 2025 – Present

University of Toronto

- **MIE429 Machine Intelligence Capstone Design** (Fall 2025, Fall 2026): mentor final-year teams building AI prototypes; 45 students in Fall 2025.
- **APS106 Fundamentals of Computer Programming** (Winter 2026): labs and tutorials in Python for 500+ students.
- **MIE1628 Cloud-Based Big Data Analytics** (Winter 2026): graduate tutorials, grading and ML feedback.

Subject Matter Expert, Corporate Training Jan 2023 – May 2025

Tata Consultancy Services (TCS), India

- Taught organization-wide programs: AI Engineer (AI ethics, NLP, computer vision), Generative AI (prompting, RAG, agents, knowledge graphs) and Data Scientist.
- Led GenAI and Python bootcamps that lifted platform learning hours by 13% in FY24.

Lead Trainer and Mentor Mar 2025 – Aug 2025

TATA ProEngage, Skill Upgradation for Rural Youth Empowerment

- Taught weekly 3-hour sessions in Python, AI and ML to 20 final-year CS/IT students over six months.

- Mentored students one-on-one on industry projects to strengthen their internship prospects.

SELECTED PROJECTS

Real-Time ASL Translator | *University of Toronto* 2026
- Led a four-person team building a deep-learning computer-vision model that translates hand signs into text.

AI Digital Twin Agent | *Personal project* 2026
- RAG and parallel tool-calling agent on Hugging Face Spaces, with prompt and cost guardrails.

Deepfake Classification and Captioning | *Tata Consultancy Services* 2024
- Designed a global hackathon challenge on CNN and Transformer models that detect and caption AI-generated images (5,000 participants).

Telecom Customer Churn during COVID-19 | *College of Engineering Trivandrum* 2021
- Machine-learning churn model (83% accuracy) whose insights informed two customer-retention schemes.

Deep Learning in Medical Imaging | *College of Engineering Trivandrum* 2020
- Seminar report on neural networks for MRI image processing and the data-availability problem.

TECHNICAL SKILLS

Programming: Python, SQL, PL/SQL, R, Java, C++
ML and evaluation: PyTorch, TensorFlow, scikit-learn, Hugging Face, QLoRA, calibration, grouped cross-validation
Agents and security: LangChain, MCP, RAG, knowledge graphs, OPA (Rego), OWASP LLM Top 10, MITRE ATLAS
Data and cloud: Pandas, NumPy, Tableau, Power BI, PostgreSQL, Docker, Git, GCP, AWS, Azure, NVIDIA DGX Spark

LEADERSHIP AND COMMUNITY

VP of Professional Development | *MIE Graduate Student Association, UofT* 2025 – Present
- Run professional development initiatives and industry-focused events for MIE graduate students.

Departmental Steward | *CUPE 3902 Unit 1, University of Toronto* 2025 – Present
- Liaison between the union and the department on professional standards and the collective agreement.

Academic Support Volunteer | *Make A Difference (MAD), India* 2019 – 2021
- Tutored 10+ children in shelter homes in Math and English, with personalized learning plans.

CERTIFICATIONS AND AWARDS

Google Generative AI Leader	2026
AWS Certified AI Practitioner	2024
Microsoft Certified: Azure Data Scientist Associate	2024
IIT Kharagpur AI4ICPS Certificate Program; IIM Bangalore Generative AI and Large Language Models	2024
AWS Certified Cloud Practitioner; PCAP Certified Associate in Python Programming	2023
TCS Digital (Hi-Talent) Cadre; TCS On the Spot Award	2022

STANDARDIZED TESTS

IELTS Academic: 8.0 overall, CEFR C1 (Listening 9.0, Reading 8.5, Writing 7.0, Speaking 7.0) Sep 2024
GRE: Quantitative 161, Verbal 150, Analytical Writing 3.5 Nov 2024